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Nanocoral Architecture for Enhanced Hydrazine Assisted Water Oxidation: Insight from Experiment and Theory

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ABSTRACT

Hydrothermally synthesized nanocoral structures of copper-cobalt sulfide is identified as a novel catalyst for electrocatalytic splitting of hydrazine (N_2H_4) in both basic and neutral mediums. Electrochemical studies in basic medium indicated that electrocatalytic splitting of hydrazine occurs at a much lower potential 0.2 V (vs Ag/AgCl) in copper-cobalt sulfide in comparison to cobalt sulfide. Gaseous analysis reveals formation of oxygen at near thermodynamic voltage of 1.23 V. Experimental observations revealed the influence of hydrazine electro-oxidation on water splitting reaction. Adsorption energy of N_2H_4 on catalyst surface and projected density of states from computational studies using Density Function Theory (DFT) proved higher activity for copper-cobalt sulfide catalyst for the electrocatalytic splitting of Hydrazine. Plausible mechanism is depicted based upon the experimental observations.

1. Introduction

Hydrazine (Hz) [N_2H_4] also named as Diazane is a prominent industrial raw material widely used as propellant for spacecrafts, corrosion inhibitors, in electrolytic plating of metals on plastic and glass, solvent for inorganic compounds and as precursor in pharmaceutical materials. Hydrazine (Hz) is preferred over other hydrocarbons for energy storage owing to the high energy density of 4269 Wh L^{-1} and 340 Wh L^{-1} in concentrated and in 10 wt% hydrazine hydrate solutions respectively. High theoretical energy density and efficiency make hydrazine a viable option for liquid fuel cells (LFC) and this concept emerged in early 1960′s itself. Electrochemical oxidation of hydrazine releases nitrogen with no catalyst poisoning species or greenhouse gases since no carbon atoms are present in it and hence hydrazine oxidation reaction (HzOR) has gained wide attraction. HzOR has emerged as an anodic oxidation reaction with faster electro-oxidation kinetics and low thermodynamic potential. In alkaline medium, standard potential for HzOR is −0.33 V vs RHE which is much lower than OER and ammonia oxidation; resulting in high theoretical cell voltage (1.56 V) in hydrazine fuel cells compared to ammonia-based cell (1.14 V) [1–3].

Hydrazine oxidation is also governed by Gerischer − Mauerer mechanism much like ammonia oxidation but without any N—N coupling reaction for the generation of nitrogen which is a major non electro-chemical rate determining step in ammonia oxidation [4]. In addition, hydrogen generation from electrochemical oxidation of hydrazine is a spontaneous reaction while in water electrolysis, significant hindrance for HER from its counter (or half-cell) OER decrease the overall splitting efficiency [5–8].

In literature, HzOR reaction has been used in combination with hydrogen evolution reaction (HER) in an attempt to reduce the oxidation potential at the anode and as a good replacement for the otherwise sluggish OER [9,10]. Low onset potential of transition metal electrodes of Ni, Co, Cu and Ag for hydrazine oxidation [11,12] has led to various studies on electro-oxidation basically on cobalt based systems. There are literature reports of HzOR in carbon paste electrode based on Co(II) complex and carbon nanotubes [13], CuO nanoflakes [14], Co_3O_4 nanowire based hydrazine sensors [15] and as bifunc-

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tional catalyst applications for HER and HzOR using CoS2 [16], NiS_2 [17,18], metallic Co [11], and Fe-doped CoS_2 nanosheets [19].

In light the of literature reports, HzOR can be perceived as an energy saving reaction which can improve the overall kinetics of the reaction it is coupled to, OER or HER. In comparison to oxides, sulfides like MnS, Co_3S_4, MoS_2, Ni_3S_2 possess enhanced conductivity, density and surface area beneficial for electrochemical studies [20–28]. Metal sulfides with multiple oxidation states are more advantageous due to multiple redox reactions and hence there is a surge towards tri/ternary metal sulfides [29,30]. Substituting copper in cobalt sulfide forms the intermediate $CuCo_2S_4$ (CCS4) compound which has been recently investigated for catalytic activity and supercapacitor applications [31–33]. CCS4 is a normal spinel comprised of trivalent cobalt (Co^{3+}) occupied at octahedral sites and divalent copper (Cu^{2+}) at the tetrahedral sites with the redox mechanism of mixed valence ions enabling the catalytic activity [34,35]. For instance, hydrothermally synthesized CCS4 has been shown to be a remarkable electrocatalyst for oxygen evolution reaction [31,33,36]. However, the utilization of such catalytically active CCS4 spinel for HzOR electrocatalysis remains unreported till now. First time, in this work, three-dimensional nanocoral CCS4 has been synthesized by hydrothermal method and is utilized as an electrocatalyst for hydrazine splitting or oxidation reaction (HzOR). We have also attempted to study the reaction kinetics and efficiency of catalyst for HzOR in basic and alkali mediums in comparison to end sulfides, Cu_2S and Co_3S_4. Electrochemical studies followed by Gas chromatography has been employed to analyze the gases evolved and to decipher the correlation of HzOR and water splitting kinetics, which is a first of its kind not reported elsewhere.

2. Experimental studies

2.1. Synthesis

Briefly, for the synthesis of CCS4, 0.2 mol of Co $(NO_3)_2{\cdot}6H_2O$ and 0.1 mol of Cu $(NO_3)_2{\cdot}3H_2O$ were dissolved in 30 mL of distilled water with vigorous stirring followed by the addition of 0.4 mol thiourea (CH_4N_2S). Two mL ethylene glycol [in contrast to hydrothermal synthesis for $CuCo_2S_4$ reported elsewhere [33] was added to this pink coloured solution and kept under vigorous stirring for 15 min. The whole solution was then transferred to a 60 mL capacity Teflon-lined stainless autoclave and kept at 180 °C for 18 h. The black coloured precipitate obtained was then washed 6 to 8 times using distilled water. This forms the $CuCo_2S_4$ washed sample (CCS4). For a comparative study, especially for electrochemical HzOR, the end members which are individual Cu and Co sulfides were also synthesized using hydrothermal method. A schematic of the synthesis procedure adopted is depicted as Scheme 1.

2.2. Instrumentation

Structural characterization was carried out using powder X-ray diffraction using Bruker D8 Discover diffractometer in the 2θ range of 20° − 70° with Cu Kα radiation of 1.54 Å. Morphology studies were performed using JEOL FE-SEM (JSM7600F) and HRTEM, FEI –TECNAI G2-20 TWIN with LaB_6 filament equipped with Bruker EDS. XPS measurements were performed using AXIS Ultra DLD spectrometer (Kratos) with a monochromatic Al Kα radiation (hυ = 1486.6 eV), as the excitation source. The XPS spectra was deconvoluted using Casa XPS program with Shirley function background correction and peak fitting using a Gaussian Lorentzian function. The recorded binding energy (BE) spectra were corrected with reference to aliphatic carbon at 285 eV.

2.3. Electrochemical studies

Electrochemical studies were performed using the conventional three-electrode system at CHI660E electrochemical workstation. All solutions were prepared using double-distilled water. To make thin slurry, a 100 mg of prepared catalyst and 200 μL of 5 % Nafion solution (as binder) was mixed followed by the addition of 300 μL of isopropanol. From this thin slurry, a measure of 3 μL was deposited on a glassy carbon working electrode (GCE). Geometric area of the electrode was 0.071 cm^2. Platinum wire/electrode was used as the counter electrode, and Ag/AgCl electrode was used as the reference electrode. Electrochemical studies for HzOR were carried out separately in two supporting electrolyte solutions like KOH (0.5 M) and K_2SO_4 (0.5 M). For cyclic voltammetry (CV) and chronoamperometric (chrono) experiments, 300 μL of hydrazine solution was taken and made up to 10 mL using 9.7 mL of KOH or K_2SO_4 supporting electrolyte.

2.4. Computational details

All the DFT calculations are done in Vienna *ab-initio* Simulation Package (VASP) code with projected augmented wave (PAW) basis set. The kinetic energy cut-off has been set to 500 eV. The exchange–correlation function has been approximated using generalized gradient approximation (GGA) of Perdew–Burke–Ernzerhof (PBE) form. The energy convergence criterion for the self-consistency loop has been set to 1×10^{-4} eV. For van der Waal corrections we have used DFT-D3 method as proposed by Grimme. Conjugate gradient (CG) algorithm has been used for the ionic relaxations until the Hellman-Feynman forces between two constituent atoms become smaller than of 0.001 eV/Å. The two-dimensional Brillouin zone has been sampled by using 5 × 5 × 1 Monkhorst-Pack k-mesh, whereas a larger k-mesh with 7 × 7 × 1 grid has been used for more accurate calculation of density of states.

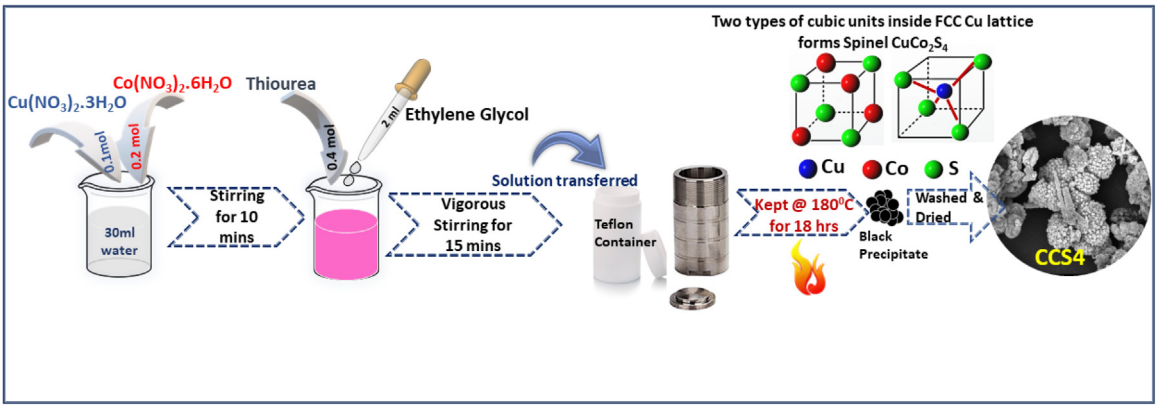
3. Results and discussions

3.1. Structure and morphology

Structural characterization using X-ray diffraction (Fig. 1(a)) showed major diffraction peaks corresponding to CCS4 (ICDD card no. 42–1450). The XRD pattern of individual Co and Cu sulfides are also shown in Fig. 1(a) and their corresponding ICDD card no.s are indicated in figure (refer S1). From FESEM, morphology of CCS4 resembles a three-dimensional nanocoral structure with average diameter of 3–4 μm shown as in Fig. 1(b). Stacking of closely packed sheet-like structures can form nanocoral structures and hence SEM morphology is an indication to sheet like structures. EDS elemental mapping confirms the presence of Cu, Co and S (Fig. 1(c–f). HR-TEM images of CCS4 (Fig. 1(g, h)) shows transparent overlapped nanosheets lying one above the other which is clearly noticed in higher magnification. Thus, each nanocoral structure (observed in SEM) is made of several close packed sheet-like structures in nano range stacked one over the other.

3.2. Electrochemical behavior of CCS4

Cyclic voltammograms (CV) of CCS4 catalyst coated GCE in 0.5 M KOH in the potential range of −0.8 to + 0.8 V at a scan rate of 50 mV s^{-1} is shown in Fig. 2(a). Profound anodic and cathodic redox peaks appeared at potential of −0.03 V/ −0.47 V(A1/C1) and 0.378 V/0.168 V (A2/C2) corresponds to Cu^{2+}/Cu^0 and Co^{3+}/Co^{2+} redox reactions. Detailed CV analysis in CCS4 and the redox mechanism using XPS is described elsewhere [36]. Upon continuous cycling, the current density varied from initial value (~40 mA cm^{-2}) and



Scheme 1. Schematic representation for the synthesis of CCS4 via Hydrothermal method.

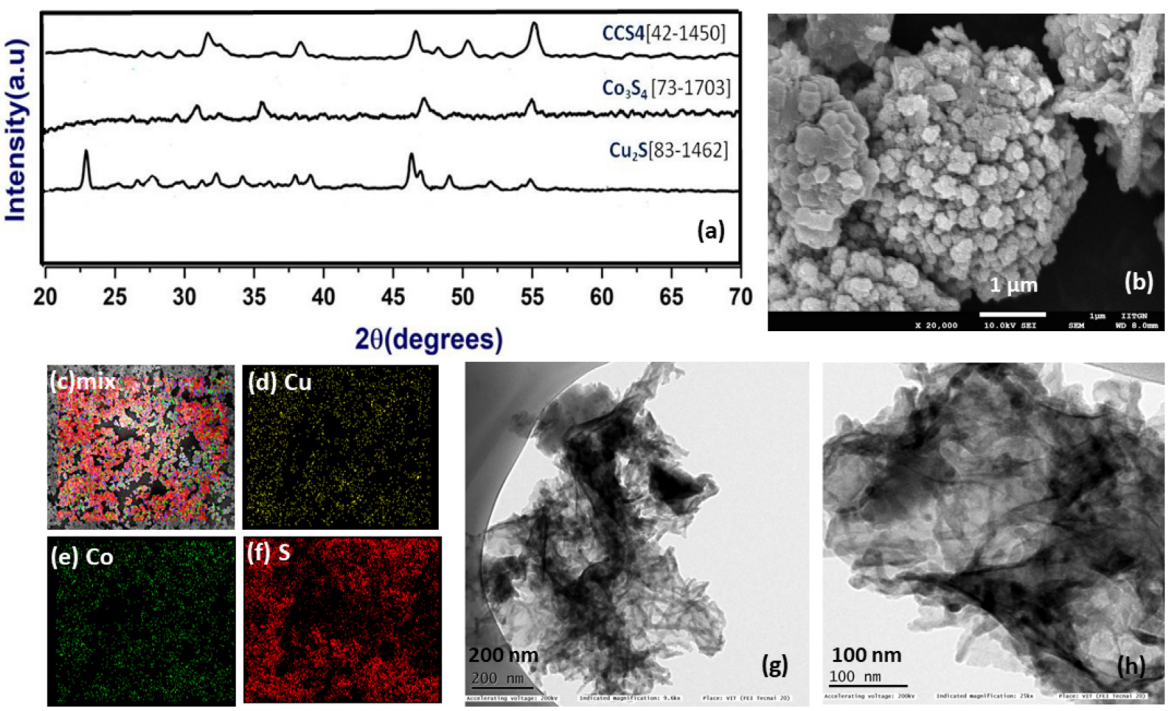


Fig. 1. Characterizations: Comparative XRD for CCS4, Co_3S_4 and Cu_2S catalysts (a), FE-SEM image of nanocoral CCS4 (b) SEM- EDS elemental mapping of CCS4 (c-f) and HR-TEM images of CCS4 (g,h). [Reprinted from [N. Sethulakshmi et al., Journal of Energy Storage 52 (2022) 104912] Copyright (2022), with permission from Elsevier]

attained a stable response (~60 mA cm^{-2}) after 20 cycles. It is noteworthy that the observable shift of ~40 mV (A1) and ~12 mV (C1) redox potential indicates overlapping of multiple redox reactions. Probably oxidation of Co^{3+} and reduction of Cu^{2+} has predominantly occurred without changing their redox-chemistry as per the Eq. (1):

$$CuCo_2S_4 \equiv Cu_1^{2+}Co_2^{3+}S_4 \leftrightarrow Cu_{1-x}^{2+}Cu_x^{+}Co_{2-y}^{3+}Co_y^{2+}S_4 \quad (1)$$

Electrocatalytic performance of the synthesized catalyst CCS4 was assessed by performing HzOR within the potential range of 0–1.0 V in 0.5 M KOH electrolyte with and without hydrazine. Fig. 2(b) represent the CV response for the electrocatalysis of hydrazine on CCS4 catalyst with an onset oxidation potential of 0.0 V (blue curve). Blank denotes the CV responses of CCS4 in KOH electrolyte i.e., in absence of hydrazine where the rise in current is noticed above 0.3 V possibly due to surface oxidation. It is interesting that CCS4 showed a very interesting response for hydrazine, with an oxidation current density of ~23 mA cm^{-2} close to 0.0 V. In the forward anodic scan, a well-defined oxidation peak at ~0.19 V with a current density value close to 40 mA cm^{-2} (Fig. 2(b), related to the oxidation of hydrazine on the CCS4 catalyst surface is observed. Fig. 2(c) and (d) show CV response of Co_3S_4 and Cu_2S for HzOR respectively.

In the case of Co_3S_4, the oxidation peak appeared at much higher potential ~0.72 V than CCS4 possibly due to higher activation energy. At this higher voltage the current density of 100 mA cm^{-2} must have significant contribution from OER along with the hydrazine oxidation because this catalyst is a well-known OER catalyst [37–39]. Cu_2S shows no response towards HzOR and is an inactive catalyst.

To examine the steady state current responses, we carried out the chronoamperometric experiment at 0.2 V and also at a higher potential 0.4 V for 200 s for the catalyst CCS4 along with hydrothermally synthesized Cu_2S and Co_3S_4 catalysts. Fig. 2(e) shows the chrono studies for CCS4 in KOH at 0.4 V where the current density gradually decreased from 80 mA cm^{-2} and quickly attain the steady state with a current density value close to 30 mA cm^{-2} for CCS4, 6 mA cm^{-2}

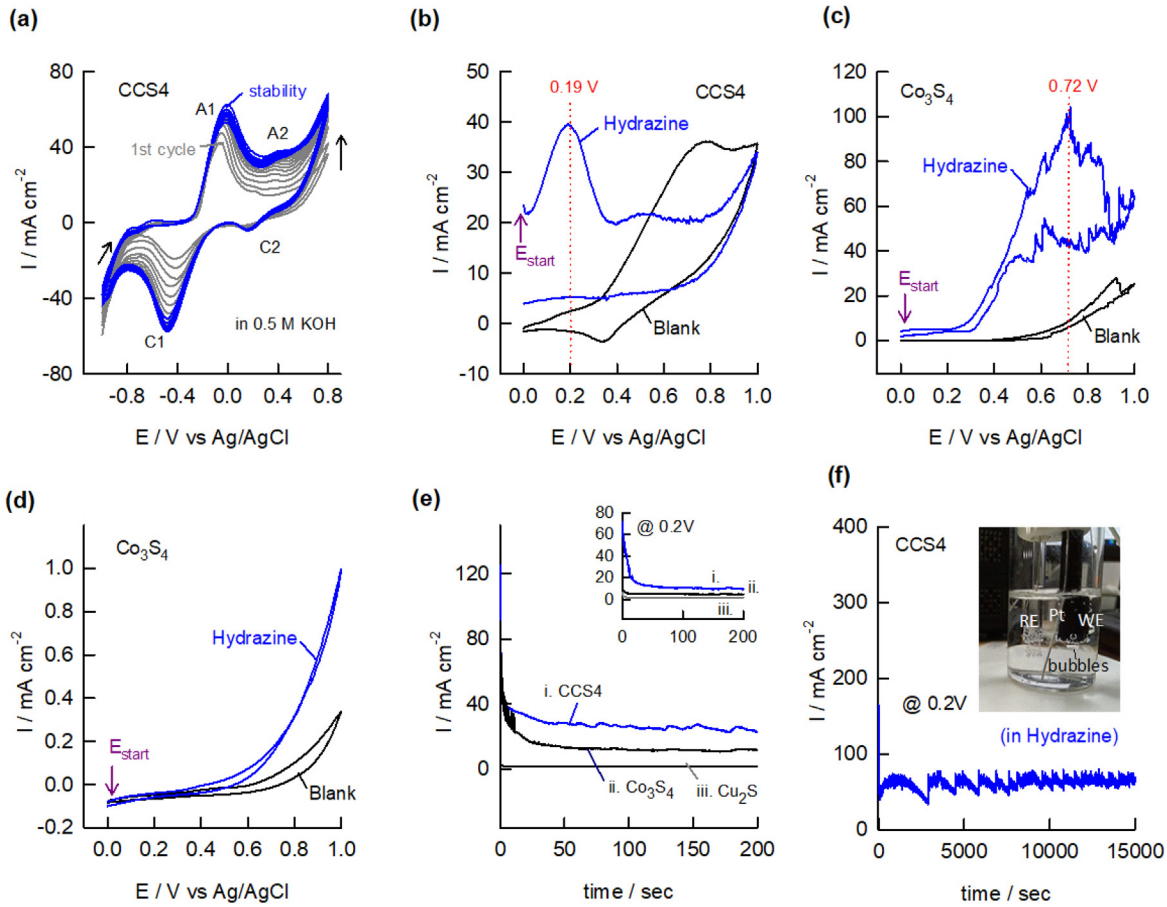


Fig. 2. Cyclic voltammetric responses of CCS4 deposited on glassy carbon electrode in 0.5 M KOH electrolyte at a scan rate of 50 mV s^{-1} (a). Electrocatalytic oxidation of 300 μL hydrazine hydrate on CCS4 catalyst (b), Co_3S_4 catalyst (c), Cu_2S catalyst (d) at a scan rate of 20 mV s^{-1} and chronoamperometric responses for all catalysts in 0.5 M KOH electrolyte at a fixed potential of 0.4 V (e) and at 0.2 V (inset). Long term stability of CCS4 for HzOR and inset image is taken during the experiment (f).

for Co_3S_4 mA cm^{-2} and 1 mA cm^{-2} for Cu_2S, respectively. Similarly, chrono at 0.2 V showed steady current densities of 11 mA cm^{-2} for CCS4, 6 mA cm^{-2} for Co_3S_4 mA cm^{-2} and 1 mA cm^{-2} for Cu_2S (inset of Fig. 2(e)). This observation clearly indicates that CCS4 has showed excellent catalytic activity towards HzOR compared to the individual sulfides. The long-term stability of CCS4 was tested using chrono studies at 0.2 V for 15,000 s (with 10 μL CCS4 catalyst coating) which showed a stable current density close to 30 mA cm^{-2} as in Fig. 2(f). In case of CCS4, another major observation is that once the potential is applied, large number of bubbles emerges out from CCS4 deposited working electrode (inset of Fig. 2(f)).

Likewise, similar CV studies were repeated for all the three catalysts, CCS4, Co_3S_4 and Cu_2S in the neutral electrolyte, 0.5 M K_2SO_4 in the potential range of −0.8 to + 0.8 V at a scan rate of 50 mV s^{-1} as shown in Fig. S2(a). Neutral medium will be susceptible to local pH changes and will show usual shifting in the potentials. In 0.5 M K_2SO_4, the redox peaks corresponding to Co and Cu species appear at shifted potentials, A1/C1 at 0.52 V / −0.05 V and A2/C2 at −0.58 V / −0.33 V as shown in Fig. S2(a). The electrocatalytic performance of CCS4 for HZOR in 0.5 M K_2SO_4 electrolyte showed a similar response as in KOH with an oxidation current density of ~ 20 mA cm^{-2} at the onset potential of 0.0 V. The HzOR oxidation peak in K_2SO_4 is observed at ~ 0.32 V with a current density value of 40 mA cm^{-2} (Fig. S2(b)) while for Co_3S_4, it is observed at ~ 0.68 V (Fig. S2(c)). No HzOR activity is observed in Cu_2S in K_2SO_4 medium as well (Fig. S2(d)). The chrono studies also followed same trend as in KOH with current density higher for CCS4 catalyst (~20 mA cm^{-2}) than Co_3S_4 (~10 mA cm^{-2}) and Cu_2S (~1 mA cm^{-2}) (Fig. S2(e). Long-term chrono studies also proved the stability of CCS4 in K_2SO_4 medium also with stable current density of 20 mA cm^{-2} for 15,000 s (Fig. S2(f)). Thus, despite the local pH effect, CCS4 is still the superior catalyst in comparison to the individual sulfides.

Electrochemical measurements were also performed in PBS solution (pH = 7.2) with hydrazine addition. Here, the current density values are lower than the two other mediums with the best activity for CCS4 catalyst (Fig. S3(a–c)). Chrono studies in PBS solution showed stable response but with much lower current densities for all the three catalysts in comparison to KOH and K_2SO_4 mediums (Fig. S3(d)). From our investigations, the ternary sulfide catalyst, CCS4 showed a stable catalytic response for HzOR than Co_3S_4 in basic, neutral and buffered mediums.

3.3. Product gas analysis

Prior to gaseous product analysis, chronoamperometric studies were carried out in an airtight electrochemical cell (Redox.me electrochemical cell (2-CEC 50 mL)). Gaseous product was collected by airlock syringe (Hamilton, 1000 μL) from the headspace of electrochemical cell after chronoamperometry and injected into the Gas chromatograph (GC; CIC Baroda). In Gas chromatograph, argon acts as the carrier gas, Haysep-A column used for separation and a thermal Conductivity Detector (TCD) for gas detection. A standard gas mixture was used for calibrating the GC results. By combining the chronoamperometry and Gas chromatography, an attempt was made to under-

stand the gas evolution and the possible reaction mechanism. In order to get significant amount of gases, 0.5 M KOH is chosen for this study. This medium has shown the highest current density among neutral and buffered medium and is free from any local pH effect. The electrochemical cell containing 0.5 M KOH solution is first purged by argon for 20 min followed by the addition of 0.5 mL hydrazine. This is followed by the chronoamperometry for 500 s at HzOR oxidation potential of 0.2 V (refer Fig. 2(b)). In GC, presence of oxygen and nitrogen are looked for in the gas collected in the headspace of the electrochemical cell.

Hydrazine dissociation in KOH medium in the presence of OH^- ions can be written as:

Anodic reaction:

$$N_2H_4 + 4OH^- \rightarrow N_2 \uparrow + 4H_2O + 4e^- \quad (2)$$

$$2H_2O \rightarrow O_2 \uparrow + 4H^+ + 4e^- \quad (3)$$

Overall anodic reaction:

$$N_2H_4 + 4OH^- \rightarrow N_2 \uparrow + 2O_2 \uparrow + 8H^+ + 12e^- \quad (4)$$

Cathodic reaction:

$$4H^+ + 4e^- \rightarrow 2H_2 \uparrow \quad (5)$$

Bar diagram in Fig. 3 and the corresponding table depicts the amount (in micromoles) of oxygen and nitrogen detected in 1 mL syringe for the two catalysts, CCS4 and Co_3S_4 after chronoamperometric measurement for 500 s on hydrazine addition in the electrolyte. Considering the volume of head space of electrochemical cell to be around 15 mL, the no. of moles of nitrogen or oxygen detected in GC after chrono of 500 s is 1/15 th of the total volume and the bar diagram in Fig. 3 denotes these values. Moles represented thus merely points towards a comparative analysis of nitrogen and oxygen gas analysis rather than giving a quantitative picture. It is to be noted that nitrogen and oxygen amount is more in CCS4 (10.4 and 21.4) than in Co_3S_4 (0.35 and 0.12) and the rate of gas production (in micromoles/s/mg) is depicted in Table 2. Clearly, the rates with CCS4 are many fold higher than Co_3S_4. Hydrogen is also detected in the counter side (not shown). Another major observation is that in the case of CCS4, OER mechanism involving oxygen production had an overpotential of 368 mV (corresponding to a potential of 0.6 V (vs Ag/AgCl) in KOH medium [Fig. S4] while on hydrazine addition in KOH, the HzOR assisted oxygen evolution occurred at a much-reduced onset potential of 0.2 V (vs Ag/AgCl) [Figs. 2 and 3]. The current density of 10 mA.cm^{-2} corresponds to a potential of 1.29 V (vs RHE), very near to the thermodynamic voltage and the overpotential for OER reduced to a lower value of 62 mV. Thus, with hydrazine, OER can be made to occur at or near thermodynamic voltage. Table 1 shows overpotentials for different catalysts and their improved values after modification of the catalyst and in our case no catalyst modification was carried out, but OER improvement is through HzOR assistance.

3.4. Plausible mechanism

Based on the information from the electrochemical studies and the gas analysis, an assumption on the reaction mechanism in KOH could be made. In KOH, OH^- ions are responsible to dissociate hydrazine molecule. Hydroxyls react with hydrazine to produce N_2, O_2 on the anode, enabling OER as a side reaction along with the hydrazine oxidation. Interestingly, OER is activated at near thermodynamic potential, 1.23 V. Thus, Hydrazine addition in KOH favours the kinetics of OER and as a result, OER occurs at a reduced potential.

Importantly, GC shows that there are about 67 % oxygen and 33 % is nitrogen in the oxidation product. This is an interesting behavior which suggests that there are more hydroxyls molecules available to react with a single hydrazine molecule over the CCS4 surface. Quantitatively, there are about 21.4 Ăμmol of O_2 and 10.4 Ăμmol of N_2 in the

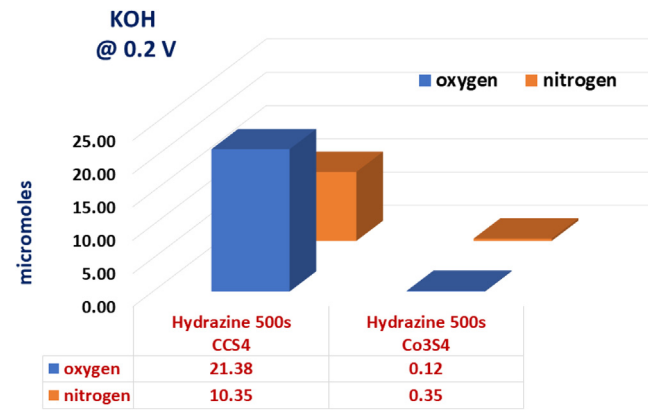


Fig. 3. Product gas analysis carried out after chrono studies for 500 s on hydrazine addition. (a) Bar diagram showing the no. of moles of nitrogen and oxygen for the catalysts, CCS4 and Co_3S_4 with tabular data in 0.5 M KOH.

oxidation product (Fig. 3) giving a ratio of 2:1 approximately. Opposite to this, estimated micromoles by faradaic efficiency are quite low, hugely underestimating the product amount. Thus, it is certain that hydrazine oxidation is occurring via chemical route as well. This is confirmed by adding CCS4 in the mixture of hydrazine and KOH followed by the analysis of the gaseous product. Interestingly, the formation of hydrogen, oxygen and nitrogen are detected, asserting the role of a heterogeneous chemical reaction:

$$NH_2.NH_2 + 2H_2O \overset{CCS4}{\rightarrow} N_2 \uparrow + 4H_2 \uparrow + O_2 \uparrow \quad (6)$$

The mechanism of this chemical reaction appear similar to what predicted by Gerischer − Mauerer in the ammonia oxidation [4] where N_2 formation occurs via NH_x intermediates. We believe, as hydrazine has $-NH_2$ species, formation of N_2 is a fast process. Interestingly, the presence of oxygen which has not been foreseen in the Gerischer − Mauerer mechanism, can be verified experimentally. Further, in the case of Co_3S_4, nitrogen and oxygen amount detected is less in KOH verifying the low activity of Co_3S_4 compared to CCS4. (Fig. 2 (c) and Fig. 3).

3.5. Post characterization studies

In order to confirm the stability of CCS4, pre- and post-XPS characterization and post, SEM-EDS Elemental mapping were also performed as shown in Fig. 4. On analyzing the XPS Cu 2p, Co 2p, S 2p spectra of CCS4 before (Fig. 4(a(i)) and after HzOR (Fig. 4(a(ii)), the peaks at binding energies of 934.4 eV and 932.5 eV corresponds to Cu^{2+} and Cu^+ species, 780.2 eV and 782.3 eV represents Co^{3+} and Co^{2+} species and peaks at 168.8 eV, 162.1 eV and 163.1 eV are oxide and metal-bonded S species, respectively [32,45]. The major observations from the XPS analyses are: (1) conversion of Cu^{2+} to Cu^+ after HzOR i.e., reduction of Cu^{2+}, (2) Conversion of Co^{2+} to Co^{3+} after HzOR or oxidation of Co^{2+} and (3) oxidation/hydroxylation post OER occurs in S 2p as well but in XPS, no specific change other than the intensity of deconvoluted peaks is observed. Thus, XPS analysis gives an evidence of multiple redox reaction of $Cu^{2+/+}$ and $Co^{3+/2+}$ in CCS4 which involves HzOR without any changes in redox chemistry and dissociation of its structure or geometry and it agrees with the proposed redox mechanism.

Considering the redox reaction on CCS4 surface, Overall HzOR reaction in CCS4 can be written as:

$$Cu^{2+}_{1-x}Cu^{+}_{x}Co^{3+}_{2-y}Co^{2+}_{y}S_4 + N_2H_4 + 4OH^- \rightarrow Cu^{2+}_{1-x-\delta}Cu^{+}_{x+\delta}Co^{3+}_{2-y+\delta'}Co^{2+}_{y-\delta'}S_4 + N_2 \uparrow + 2O_2 \uparrow + 8H^+ + 12e^- \quad (7)$$

Table 1
shows OER overpotentials for different catalysts and the improved potentials after modification. [potentials shown in bold are improved potentials].

Catalyst	Electrolyte	Current density mA.cm^{-2}	Overpotential (η) mV	Technique for modification of OER	Reference
V2O5	1 M KOH	10	1330	Vanadium doped Co/CoO ecapsulated in carbon nanorod	[40]
Co/C			431		
V-Co/CoO @C			320		
VN-Co	1 M KOH	10	395	Co and P doping in Vanadium nitride nanowires	[41]
VN-P			401		
Vn-Co-P			335		
Pure Mo_2C	1 M KOH	20	380	Zr based metal organic framework(MOF)[Uio-66] / Mo_2C hybrids	[42]
Mo2C/Uio-66 hybrid			180		
(Co,Fe)OOH	1M KOH	10	258	Sulfur incorporation in Co,Fe –(oxy/hydroxide)	[43]
S-(Co,Fe)OOH			240		
Pristine GOT	1M KOH	10	460	Graphene oxide functionalization with Polyethyleneimine (P-GOT) and ethylenediamine(E-GOT)	[44]
E-GOT			310		
P-GOT			240		
CCS4	0.5M KOH	10	368	Hydrazine addition in KOH, HzOR assisted OER	[36]
Co_3S_4			500		
CCS4			62		This work
Co_3S_4			368		

Table 2
Rate of gas production.

Rate of gas production (micromoles/s/mg)		
Catalyst	**Oxygen**	**Nitrogen**
CCS4	1.069	0.5175
Co_3S_4	0.006	0.0175

Further, in the SEM image of post characterized CCS4 sample (Fig. 4(b)), morphology is coral like structure as observed in the synthesized sample (Fig. 1(b) indicating that morphology is maintained significantly post HzOR. The EDS elemental mapping (Fig. 4(c-f)) shows the the presence of Cu, Co and S in CCS4 confirming that no significant dissolution of metal ions from the catalyst surface takes place after the electrochemical studies and HzOR. Thus post characterization studies, SEM, EDS mapping and XPS supports the stability of the CCS4 catalyst post HzOR.

4. Theoretical analysis based on density functional Theory (DFT) calculations

In order to validate the experimental findings of hydrazine (N_2H_4) splitting, we have performed a first-principles based density functional Theory (DFT) calculations. Theoretically, the ability of a surface to catalyze N_2H_4 splitting is determined by the adsorption energy of N_2H_4*, NH_3* and N* adsorbed on the catalyst surface. Henceforth, this theoretical study primarily focuses on the adsorption characteristics of the above mentioned three adsorbates, on top of Co_3S_4 and CCS4 surfaces. Additionally, we have also investigated adsorption of O and OH to envisage oxygen evolution reaction (OER) on CCS4 surface.

The Co_3S_4 surface structure has been simulated from bulk Co_3S_4, by using a slab model with a vacuum of ~ 15 A along (0 0 1) direction that contains 24 Co atoms and 32 S atoms as shown in Fig. 5. As we can see from the figure that there could be two possible surface termination of Co_3S_4 surface. However, throughout our calculation, we have used the

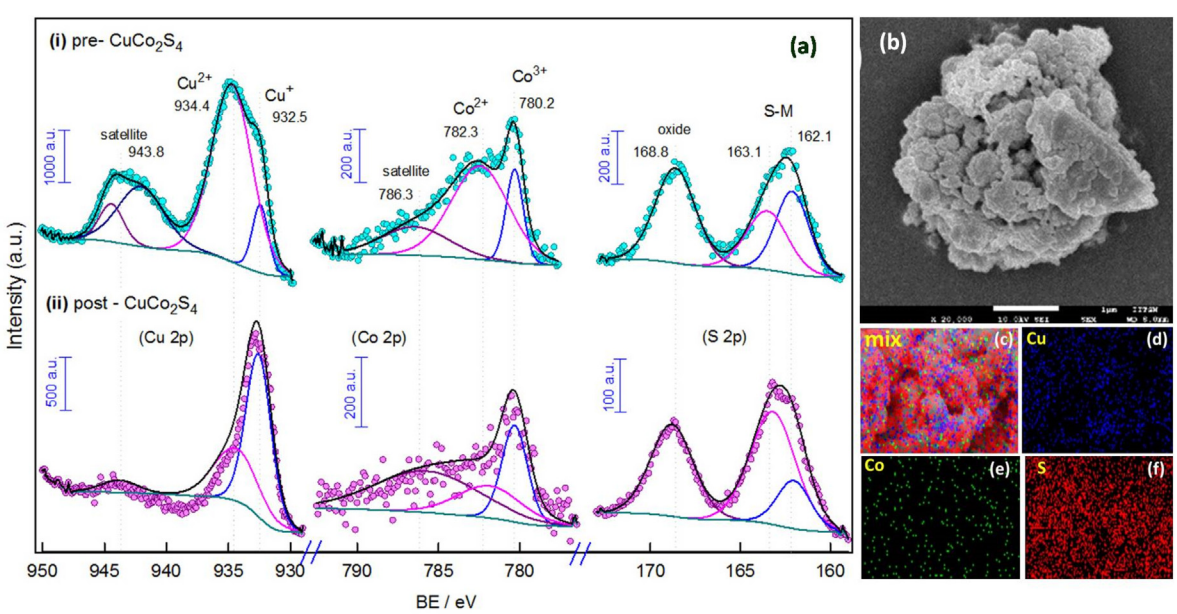


Fig. 4. Post characterizations of CCS4 catalyst after HzOR: (a) high resolution XPS spectra of CCS4 showing Cu $2p_{3/2}$, Co $2p_{3/2}$ and S 2p before (i) and after (ii) chronoamperometric responses at a fixed potential of 0.4 V for 1000 *sec* in 300 μL hydrazine hydrate 0.5 M K_2SO_4 electrolyte. (b) Post-SEM image along with respective EDS elemental mapping (c-f).

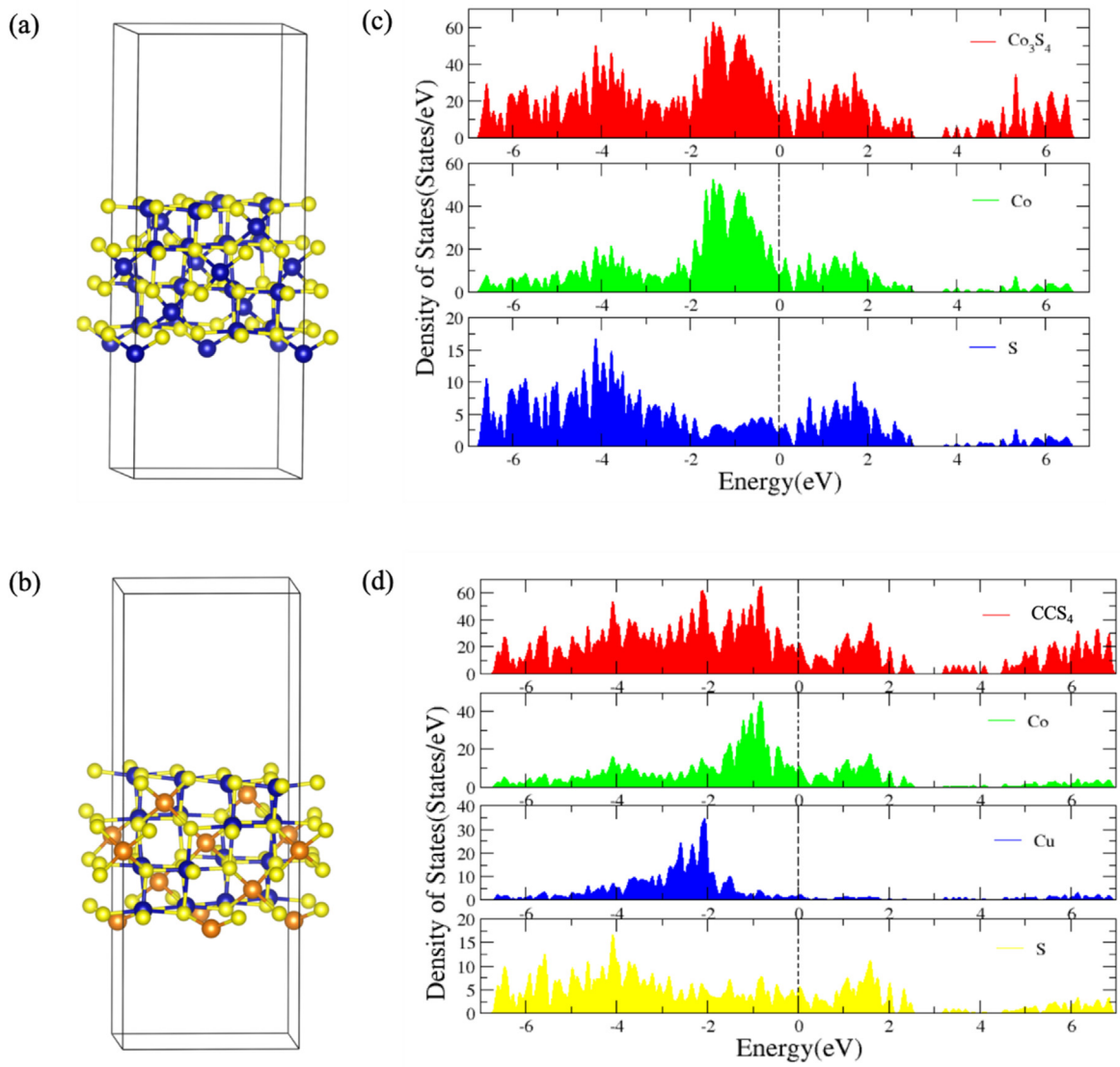


Fig. 5. Optimized geometries and partial density of states plot for: (a) Co_3S_4 surface; (b) CCS4 pdos; (c) Co_3S_4 pdos and (d) CCS4 pdos. Blue, orange and yellow balls represent Co, Cu and S atoms respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Adsorption energy (E_{ads}) of N_2H_4*, NH_3* and N* on top of Co_3S_4 and CCS4 surfaces.

Surface	E^{N}_{ads}(eV)	$E^{NH_3}_{ads}$(eV)	$E^{N_2H_4}_{ads}$(eV)
Co_3S_4	−4.81	−1.05	−1.1
CCS4	−4.11	−1.09	−1.17

one with CoS termination as it is energetically more preferable under S-rich condition. In addition, this particular termination provides both Co and S sites for the possible adsorption of N_2H_4*, NH_3* and N*. In the next step CCS4 surface has been modelled by replacing 8 Co atoms been by Cu which with a doping concentration of ~ 33.33 %.

In order to get a more detailed picture of how Cu doping influences the electronic and surface structures of Co_3S_4 we have analyzed their partial density of states (pdos) (See Fig. 5) and also determined their work function. However, from the pdos plot it has been clear that doping does not cause a huge change in the electronic property of Co_3S_4 systems as both the systems are metallic in nature. The Cu projected pdos plot (Fig. 5(d)) shows that the contribution from Cu atoms are mainly prominent in the valence band region.

The adsorption energy for different adsorbates during hydrogen production is summarized in Table 3 and the reaction coordinate is depicted in Fig. 6. From the reaction coordinate plot, it is clear that the step from N_2H_4* →NH_3* formation is uphill in energy diagram while the last one for the formation of NH_3* →N* downhill indicating the last step is exothermic. Now the nitrogen adsorption energy values on both CCS4 and Co_3S_4 surfaces are large negative suggesting the binding is of strong chemisorption type. The last step is rate determining step here and if we look at the barrier of reaction it is less in case of CCS4 surface. Hence the reaction is much likely to occur on CCS4 as compared to the Co_3S_4 surface. All of these findings lead to the conclusion that the reaction will occur more spontaneously on CCS4 catalyst surface as compared to the pristine Co_3S_4 one. This observation matches well with the experimental findings.

In the next step we have calculated O (E^{O}_{ads}) and OH (E^{OH}_{ads}) adsorption energy on top of CCS4 catalyst surface which turns out to be −1.76 eV and −2.76 eV respectively. Both these values represent adsorption energy within the chemisorption range. From the values it is clear that E^{OH}_{ads} is more negative compared to E^{O}_{ads} which is a signature of OER mechanism.

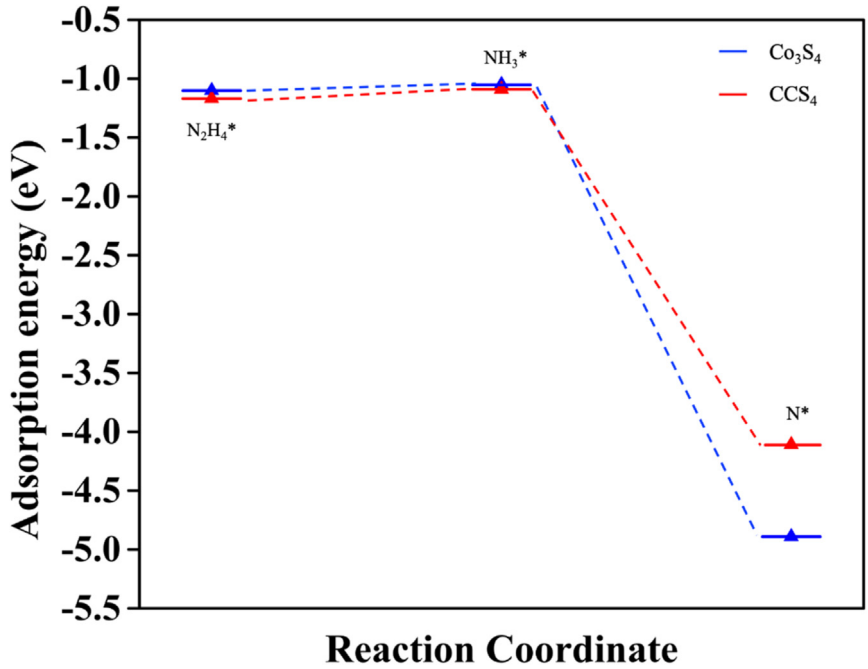


Fig. 6. Reaction coordinate mapping for N_2H_4*, NH_3* and N* adsorption on top of Co_3S_4 and CCS4 surfaces.

5. Conclusion

In summary, hydrothermally synthesized nanocoral CCS4 spinel has been utilized as an efficient and stable electrocatalyst for HzOR in both alkaline and neutral electrolytes. XRD and microscopic characterizations confirm the formation of nanocoral structures. Electrochemical studies in KOH, K_2SO_4 and neutral buffer mediums revealed that electro-oxidation of hydrazine occur at a lower potential (0.2 $V_{Ag/AgCl}$) with highest current density (30 mA cm^{-2}) in CCS4 catalyst. The catalytic activities of individual sulfides (Co_3S_4, Cu_2S) are comparatively lower or inactive for this reaction in all medium. Gas chromatography studies indicated the presence of nitrogen, oxygen in the oxidation product and hydrogen as the counter product in CCS4. It is worth mentioning that hydrazine oxidation takes place via chemical as well as electrochemical route as the amount of product outweigh the product estimated by faradaic efficiency. Hydrazine also assists the oxygen evolution reaction (OER) to happen at near thermodynamic voltage (1.29 V (vs RHE)), with the involvement of hydroxyl ions in KOH medium. XPS studies revealed that redox interaction of Co and Cu ions in CCS4 catalyst accounts for its superior electrochemical activity. DFT studies to determine the absorption characteristics of N_2H_4 showed better catalytic performance for CCS4 than Co_3S_4. PDOS revealed that presence of Cu in CCS4 greatly influences the electronic structure and properties. CCS4 is a stable catalyst and show no substantial change in the morphology or chemical composition post electrocatalytic HzOR as revealed from post electrocatalytic characterizations like XPS and SEM-EDS.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jelechem.2022.116776.

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